

SISC: Sensor-based Intelligent Shopping Cart

Rajlakshmi Badi

Department of Computer Science and Engineering
Walchand College of Engineering
Sangli, India
lakshmikatyare@yahoo.com

Bashirahamad Momin

Department of Computer Science and Engineering
Walchand College of Engineering
Sangli, India
bfmomin@yahoo.com

Abstract— The most inconvenient part in the shopping mall is waiting in the queues and lack of product information or sales, which consumes more time at billing counters. The proposed system SISC will give the customer a quick shopping experience by providing product sales promotion on the customer's smartphone using machine learning algorithms, and update total bill amount on a display screen attached to cart using RFID technology. This will make sure to the customer, shopping is under budget or over budget and also direct promotions on a smartphone will arouse his/her purchasing desire. Also, this system will reduce the supermarket budget invested in the manpower.

Keywords—Eclat, Frequent Item Set Mining, Machine to Machine, RFID Technology

I. INTRODUCTION

Recently, the extensive utilization of embedded systems, inexpensive sensors, and wireless sensor network has fostered the rise of Industrial Internet of Things (IIoT). It alludes to the integration of intricate devices and objects with high-end applications or software and networked devices and sensors. The IIoT empowers businesses to analyze value or data after collecting and aggregating them from sensors attached to devices. It maximizes machine's efficiency and entire operation's throughput [1]. Applications of IIoT include predictive maintenance, smart grid, motion control, big data analytics, smart medicine, and M2M [2].

M2M communication system lets the machines and devices to exchange data or sensor values with each other independently [3]. The M2M system includes following key components, they are, devices attached to sensors or actuators, RFID technology, wireless network and autonomic computing systems, programmed to service networked devices to interpret data and make decisions [4]. Applying the M2M System to supermarkets and malls is an attractive direction for customers as well as for employees.

Supermarket or retail system provides all daily necessities to people ranging from groceries, food products, electrical appliances, clothing, etc. Sometimes customers face many problems while shopping such as incomplete information about discounts on product, worrying that amount of money carried is sufficient or not, and waiting for a long time in queues for billing at billing counters is a tedious task while shopping. Endless progress is required in the existing billing system to achieve more quality in retail systems.

RFID technology is one of the most innovative technologies combining it with the M2M system will foster the ubiquitous retail sales. Also, this technology offers a flexible system for shopping malls, which improves the shopping experience for the customers in any self-service retail store. As

this new inexpensive and small-sized RFID tag developing techniques have emerged, RFID technology is increasingly encouraging to replace the barcode system [6].

In Automatic billing systems, RFID technology is used to solve the problem of waiting for a long time in the queue at billing counters of supermarkets and saves the time of the customer as well as the cashier.

II. LITERATURE SURVEY

A. Machine-to-Machine (M2M) communication

For future generation communication systems, an M2M communication system is the most promising technology. This communication system provides a universal communications system with full computerized machines. These machines and devices are largely connected to each other using wired or wireless links, without the direct involvement of human [3].

B. Sales Promotion based on Behavior of Customer [5]

This application comprises the data collection by sensing values from the shopping cart and taking appropriate decisions by the server. This system helps the customer to select the products and thus arouses customer's purchasing desire. To allow the customers to easily use this system, [5] adopted two ordinary actions or behaviors of a customer regarding the handling of the cart that observes his/her behavior: (a) Holding the handle of the cart, and (b) motion of the cart. These two actions are combinedly observed and used by the author of [5] to infer the nine possible cases which will be used to analyze the customer behavior. Those behavioral results include having no, moderate or significant interest in the product. Also, whether the customer requires any assistance or not, depending on the number of times he visits the same shelf. In addition, to save energy of the cart's battery, the cart's system is put into standby mode, depending on the time the cart is idle or in use.

C. Association Rule Mining (ARM)

Author of paper [12] has discussed many ARM algorithms that are composed of following characteristics: (a) Design Approach such as, bottom-up approach, top-down and hybrid approaches, (b) Candidate Generation, (c) Layout: Two types of layouts are mainly used in Frequent Itemset Mining: Horizontal and Vertical, (d) Interestingness Measure: There are two Interestingness measures: Descriptive measure and Probabilistic measure, (e) Database: In Database, data are classified into certain and uncertain. The certain database includes data in the form of transactional, unstructured and

semi-structured. The uncertain database includes data like GPS, Sensor values, Weather etc. (f) Frequent Itemsets.

D. Frequent Itemset Mining [8], [9]

To mine association rules efficiently, several techniques have proposed to generate frequent itemsets. The methodologies of generating frequent itemsets are divided into basic three procedures [13].

- 1) Horizontal layout: For eg. Apriori algorithm.
- 2) Vertical layout: For eg. Eclat algorithm.
- 3) Projected database: For eg. FP- Growth algorithm.

1) *Apriori*: This algorithm of [8] follows breadth-first search procedure where the traversing of the tree is processed one level in a single pass. Apriori works with a two-pass approach for identifying frequent itemsets. This approach is used for all levels. In the first scan, a list containing candidate itemsets is generated and supersets of infrequent remaining itemsets are pruned at each level. In the second scan, the support values are calculated for the itemsets that are pruned from first scan and further those itemsets having a support below the threshold defined by the user are pruned. There are two ways to calculate support for all the itemsets. (a) for each itemset, by counting the total number of transactions present in the database or as an alternative by traversing all the transactions, (b) or by counting the itemsets belonging to each transaction.

2) *Eclat*: Equivalence Class Clustering and bottom-up Lattice Traversal (Eclat) follows depth-first search procedure where itemsets having identical prefix are traversed at first. The algorithm initiates at level one and starts with leftmost node. Eclat traverse tree in a downward direction and then prunes itemsets having support less than the threshold given by the user. When all nodes having the same prefix have been processed, then this algorithm backtracks and starts with the next level. Then, at the next level, Eclat continues processing the remaining itemsets having the different prefix. Eclat maintains a list of transactions for each item for calculating support for those itemsets [9].

3) *FP Growth*: Frequent pattern growth (FP-Growth) is implemented by using a tree to store in the form of actual baskets, instead of storing candidates like Apriori and Eclat stores. The FP-Growth's tree structure provides a vertical view of the itemset. However, a header table is added while frequent pattern growth for every single item that has support value above the threshold given as a support level. To connect all similar nodes, the header table holds a linked-list through the tree. The header table provides FP-Growth a horizontal view of the data, along with the vertical view provided by the tree [9].

Apriori is an easily comprehensible frequent itemset mining algorithm. Hence, Apriori is a well-known beginning stage for studying about frequent itemset mining. Though, compared to Eclat and FP-Growth algorithms, Apriori has serious scalability issues and exhausts existing memory more rapidly. To avoid this situation, for large datasets Apriori algorithm should not be used. Those applications having large database should use either Eclat algorithm or FP-Growth algorithm to find frequent itemsets. The FP-Growth algorithm is more complex for implementation than Eclat. However, it gives considerably better performance than Eclat [9].

E. Comparison of Barcode, QR-code and RFID System [6]

Automatic identification and data collection (AIDC), refers to numerous technologies that help to reduce labor and time and eliminate human errors by replacing manual techniques of data collection and data entry. Barcode and similar technologies such as QR code, and RFID technologies are one of the several AIDC technologies.

Barcode refers to an optical machine-readable parallel lined representation used to represent data about the product attached to it. Unlike barcode system, RFID Technology uses radio frequency, also called electromagnetic waves to transfer information from an RFID tag attached to a product or device to the RFID reader.

Quick Response (QR) codes display a small bit of information very conveniently, that is simply scanned and processed usually by smartphones. This allows physical objects to become less interactive, by providing information that is easily scanned like a website URL or any advertising material.

Also, the author has compared the all three technologies using various factors as shown in Table I. Compared to other similar technologies, RFID technology is being proved to be a fostering technology.

F. Automatic Billing System [7]

The purpose of [7] is to facilitate an automatic billing system which is implemented using two technologies namely, RFID and ZigBee technology. This system avoids the waiting time in queue for billing at billing counters of supermarkets. These supermarkets are installed with anti-theft bars at exit gates, to avoid the stealing of products.

G. Product-Navigation [5]

3S cart system of [5] splits the supermarket into many grids of the aisle, for example, product shelves for navigating the path. In this system, customer's location is given as source grid and the location of the desired product is given as a terminal grid, author of [5] used the A* algorithm to locate the briefest way amongst customer and product.

III. PROPOSED SYSTEM AND METHODOLOGY

Sensor-based Intelligent Shopping Cart, denoted by SISC, aims to provide an automatic billing system attached to the shopping cart and gives a sales promotion on the customer's smartphone. This system is implemented using RFID technology and machine learning algorithms. RFID technology is used to implement the automatic billing system, whereas frequent itemset generation algorithm is used for sales promotion.

TABLE I. COMPARISON OF BARCODE, QR CODE, AND RFID TECHNOLOGY

Attribute	RFID	Barcode	QR Code
Accessibility Range	≤ few meters	≤ few inches	≤ few inches
Line of Sight	Not Required	Required	Required
Read/Write	Read and Write	Read Only	Read Only
Information Capacity	More than QR code and Barcode	Very Less	Less
Reliability	Damaged tags can work flawlessly	Damaged tags won't work	Up to 30% damaged tags can work
Cost (in bulk)	\$0.01	\$0.001	\$0.001

A. Designing a proposed model of SISC system:

As shown in Fig. 1, a wireless router will be used for transmission of data between microcontroller and master computer. The customer will scan the required item and will add to cart. There are three buttons provided on the cart-remove, checkout, and reset. If the customer presses remove button on the cart, recent scanned item's amount is deducted from the total amount. If the customer presses the checkout button, a bill will be generated by a master computer for the respective cart. If customer presses reset button, the cart's total amount will become zero. At the checkout counter, the customer has to pay bills as per the bill generated by a master computer. Customers will use smartphones to view the sales promotion, sent by the master computer.

B. Configuring required setup for SISC system:

Figure 2 shows the hardware setup for the system. The hardware required for the SISC implementation is Arduino ATmega2560, ESP8266 Wi-Fi module, 16x2 LCD Display, MIRF522 RFID Reader, MIRF522 RFID Tags attached to the product, TTP-223 Touch keys, Wire, Battery, and the capacitor.

Figure 3 shows a conceptual view of SISC system, after deploying this system to existing shopping carts.



Fig 3: Conceptual view of SISC system, [10].

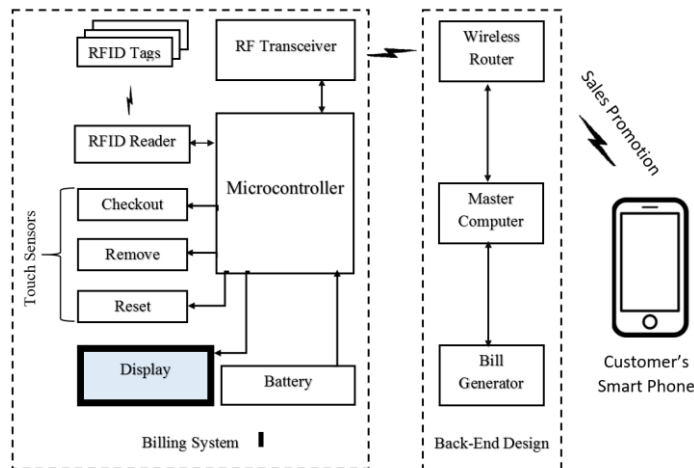


Fig 1: Block Diagram of SISC system.

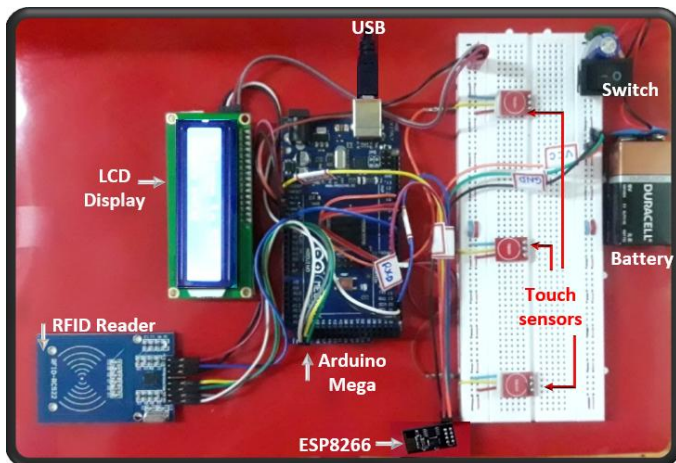


Fig 2: Hardware setup for SISC system.

C. Implementation of SISC system:

Sensor-based Intelligent Shopping Cart (SISC) is divided into two main modules: Sales Promotion on the smartphone of the customer, and Automatic Billing System (on shopping cart).

1) *Sales Promotion*: This module involves instant display (on customer's smartphone) of offers, top selling products, and recommended products based on products already dropped into the cart.

Client Side steps:

- Each cart is provided with a unique QR code sticker.
- The customer has to connect his/her smartphone to mall's Wi-Fi access point.
- The customer scans the QR code, a URL with the unique cart code is displayed on the smartphone.
- Customer opens the provided link using any browser.
- He/she will be able to see offers, top selling products, and recommended products on his/her smartphone.

Server Side (Master computer) steps:

- Run Association Rule Mining algorithm i.e. Eclat Frequent Itemset Algorithm as given in [11], on the master computer using a database of billed items per transaction, to generate frequent itemsets and rules.
- Find the missing products in the cart using those generated frequent itemsets and rules.
- Recommend those missing products to the customer on the connected smartphone.

2) *Automatic Billing System*: This module shows an instant display of updated bill on the display screen of the cart as soon as customer scans the product to add to the cart or remove from the cart. RFID tags and reader will be used to implement this module. Following is the proposed system:

- Every product in the mall will have an RFID tag on it.
- Each Cart will contain a SISC system attached to it.
- If the product is added to the cart or removed from the cart, it must update the bill on the LCD screen.
- There will be a centralized server system called master computer to process the bill.
- After the payment of money, the cart gets reset.

D. Evaluation of the SISC system:

The performance of the SISC system can be analyzed in terms of easiness to the user for fast billing, the more precise recommendations in real-time, and comparison with the existing system.

IV. CONCLUSION

According to the previous work of the Smart Cart system, product navigation, sales promotion depending on customer behavior, Automatic billing system, has been added as a feature of the smart cart. Sales promotion by recommending products in online shopping websites is normally achieved by association rule mining algorithms. This will be the first smart cart offline retail system, that will recommend products to customers based on the products dropped in the cart by customers. In this SISC system, the combination of frequent itemset generation algorithm and automatic billing system using RFID technology will give the customers ease of automatic billing and sales promotion based on their product purchase. For implementing this system in real time, barcode system needs to be replaced by RFID system.

REFERENCES

- [1] J. Wan, Z. Shu, D. Li, S. Wang, M. Imran, A. Vasilakos and S. Tang, "Software-defined Industrial Internet of Things in the context of Industry 4.0", *IEEE Sensors Journal*, vol. 16, no. 20, pp. 7373-7380.
- [2] <https://www.xilinx.com/applications/megatrends/industrial-iiot.html>
- [3] Z. Meng and Z. Wu, "A data-oriented M2M messaging mechanism for industrial IoT applications", *IEEE Internet of Things Journal*, vol. 4, no. 1, pp. 236–246, Feb. 2017.
- [4] <http://internetofthingsagenda.techtarget.com/definition/machine-to-machine-M2M>
- [5] Y. Wang, and C. Yang, "3S-cart: A lightweight, interactive sensor-based cart for smart shopping in supermarkets", *IEEE Sensors Journal*, Vol. 16, No. 17, September 1, 2016.
- [6] P. Shah, J. Jha, N. Khetera, and M. Zala, "A literature review on improving error accuracy and range based on RFID for smart shopping", *IJSRD - International Journal for Scientific Research & Development*, Vol. 3, Issue 10, 2015, ISSN (online): 2321-0613.
- [7] A. Yewatkar, F. Inamdar, R. Singh, Ayushya, A. Bandal, "Smart cart with automatic billing, product information, product recommendation using RFID & ZigBee with anti-theft", 7th International Conference on Communication, Computing, and Virtualization 2016.
- [8] A. Mahanti, and R. Alhajj, "Visual interface for online watching of frequent itemset generation in apriori and eclat", *Proceedings of the Fourth International Conference on Machine Learning and Applications (ICMLA'05) IEEE*, 2005.
- [9] J. Heaton, "Comparing dataset characteristics that favor the apriori, eclat or FP-Growth frequent itemset mining algorithms", *IEEE*, 2016.
- [10] <http://strategyonline.ca/2011/09/20/grocery-shopping-2-0-the-techno-cart/angle-2/>
- [11] B. Goethals, "Survey on frequent pattern mining," *University of Helsinki*, 2003
- [12] P. Rani and I. Ranjan, "A comprehensive study of map-reduce frequent itemset mining: a survey", *International Journal of Control Theory and Applications*, Vol. 9 Issue 46, pp.31-44, 2016.
- [13] A. Saxena, and S. Gadhiya, "A survey on frequent pattern mining methods apriori, eclat, FP growth", *International Journal of Engineering Development and Research*, Vol. 2, Issue 1, pp. 92-96, 2014.